

Deterministic pose-spline dynamics estimator

Estimator structure

pose-only continuous-time spline -> analytic derivatives
fixed-spline gradient matching -> shared physical parameters and lag
shooting nodes: none
continuity constraints: none
sensor/IMU channels in loss: none

Initial estimate

source kind: multiple_shooting_result
source path: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/output/deterministic_multiple_shooting_multi/result.json
source mass [kg]: 3.02232298411
selected initial mass [kg]: 3.02232298411

Selected strict-ZOH parameters

mass [kg]: 2.76395719961
inertia [kg m²]:
[0.0727819009231, -0.00157574128676, 0.00017021215867]
[-0.00157574128676, 0.145464959033, 0.000121049039743]
[0.00017021215867, 0.000121049039743, 0.217282275919]
principal moments [kg m²]: [0.072747548907, 0.145498912684, 0.217282674284]
inertia triangle margin [kg m²]: 0.00096378730726
CoG offset [m]: [-0.0028548317789, -0.000259983964135, -0.0162912081759]
rotor force effectiveness: [1.00294882179, 1.03446692886, 0.972415930905, 0.991180047602]
lag [s]: 0.048338405532
joint dynamics loss: 0.383758618049
soft-prior cost: 0.0456807044696

Smooth physical coordinates

log_mass_scale	0.161568497783
log_second_moment_cholesky_xx_scale	0.405188852452
log_second_moment_cholesky_yy_scale	0.0568051583329
log_second_moment_cholesky_zz_scale	0.00166042384935
normalized_second_moment_cholesky_yx_offset	0.0162839350461
normalized_second_moment_cholesky_zx_offset	-0.00498864757033
normalized_second_moment_cholesky_zy_offset	-0.00593920262519
cog_offset_x_m	-0.0028548317789
cog_offset_y_m	-0.000259983964135
cog_offset_z_m	-0.0162912081759
force_effectiveness_contrast_1	-0.0218791332027
force_effectiveness_contrast_2	0.0378748429016
force_effectiveness_contrast_3	0.0102295826074

Spline and collocation

collocation step [s]: 0.01

Bag failure_1

selected spline knot spacing [s]: 0.1
dynamics loss: 0.669190097396
estimated forward position RMSE [m]: 23.1316402934
estimated + external wrench position RMSE [m]: 2.92263162087
external wrench improves estimated free rollout: True
nominal forward position RMSE [m]: 56.2552485683

estimated improves nominal: True
residual wrench mean: [0.088042 0.078711 0.058019 -0.0298 -0.009029 -0.00958]
residual wrench RMSE: [1.122817 0.737714 1.471983 0.128496 0.632377 0.427237]
residual wrench trend/s: [0.105839 0.041043 -0.567822 0.003477 -0.017555 0.02326]
residual wrench cumulative impulse: [0.541265 0.450173 0.330023 -0.177132 -0.054916
-0.057299]

Bag failure_2

selected spline knot spacing [s]: 0.2
dynamics loss: 0.098327138702
estimated forward position RMSE [m]: 51.3112593647
estimated + external wrench position RMSE [m]: 0.875209771044
external wrench improves estimated free rollout: True
nominal forward position RMSE [m]: 50.2776651111
estimated improves nominal: False
residual wrench mean: [0.070862 -0.362079 -0.019717 0.026933 0.024305 0.01447]
residual wrench RMSE: [0.246384 0.447398 0.712196 0.04521 0.2421 0.08776]
residual wrench trend/s: [-0.039577 0.030114 -0.281661 -0.009902 0.046324 0.00336]
residual wrench cumulative impulse: [0.391955 -1.966245 -0.117245 0.146912 0.127858
0.078193]

Smoothstep continuation

width=0.5: delay=0.0476858722s, cost=0.429349362069, nfev=16
width=0.2: delay=0.0484578455s, cost=0.429467756736, nfev=10
width=0.05: delay=0.0483384055s, cost=0.429437416086, nfev=4

Strict-ZOH polish

delay=0.0443384055s: screening=0.430320116564, refined=nan, selected=False
delay=0.0453384055s: screening=0.430384233902, refined=nan, selected=False
delay=0.0463384055s: screening=0.430296654816, refined=nan, selected=False
delay=0.0473384055s: screening=0.43208906756, refined=nan, selected=False
delay=0.0483384055s: screening=0.429439324806, refined=0.429439322519, selected=True
delay=0.0493384055s: screening=0.429862691903, refined=0.429861764707, selected=False
delay=0.0503384055s: screening=0.429868798721, refined=0.429867832441, selected=False
delay=0.0513384055s: screening=0.429975216206, refined=nan, selected=False
delay=0.0523384055s: screening=0.432475568738, refined=nan, selected=False